AGRO AI — Training Datasets Specification

Complete Technical Documentation of Machine Learning & Computer Vision Datasets

Platform: AGRO AI (Qwen & Ollama Hybrid) • **Date:** November 03, 2026 • **Status:** Production Verified

Executive Summary: *This technical specification document details the complete suite of datasets engineered and utilized to train, calibrate, and benchmark the machine learning and computer vision models powering the* ***AGRO AI*** *intelligent agriculture platform. It covers 4 primary structured tabular CSV datasets (comprising* ***63,900 balanced samples*** *across* ***48 numerical feature dimensions****), underlying open-access raw vision datasets (PlantVillage, PlantDoc, Kaggle Pedological Soil, ICAR collections), the 10,208-taxa Indian botanical reference directory, and the multimodal vision-language foundation model (Qwen 2.5-VL 3B).*

1. Master Dataset Inventory & Core Metrics

AGRO AI operates a stratified multi-tier architecture consisting of edge-ready Random Forest / Gradient Boosting ensemble classifiers, an ONNX convolutional vision pipeline, an authoritative PostgreSQL botanical database, and an on-premise multimodal vision LLM:

Dataset File	Target Model / Task	Total Rows	Features	Classes	File Size
agricultural_relevance_dataset.csv	Input Gatekeeper (Relevance Filter)	9,000	48 (Color/Texture/Edge)	3 classes	2.25 MB
crop_disease_dataset.csv	Crop Foliar Pathogen Detection	31,200	48 (Pathology/Spectral)	39 classes	8.78 MB
plant_species_dataset.csv	Botanical Species Identification	14,700	48 + 4 Metadata	98 species	7.61 MB
soil_pedological_dataset.csv	Pedological Soil Classification	9,000	48 (Granulometry/Tone)	6 soil types	2.32 MB
Plant Species Master Directory	**Database Reference Catalog**	**10,208 taxa**	**16 schema attributes**	**All India Flora**	**14.5 MB**
Qwen 2.5-VL 3B Model	**Zero-Shot Visual Diagnostic LLM**	**3.0 Billion Param**	**Multimodal Vision Tensor**	**Open Vocabulary**	**1.9 GB**

2. Underlying Raw Datasets, Benchmarks & Open Licenses

The engineered feature datasets and pretrained weights originate from internationally recognized agricultural and botanical research repositories:

- **PlantVillage Dataset (CrowdAI / Penn State):** Contains 54,306 laboratory and controlled-condition photographs of healthy and infected plant leaves across 14 crop species and 38 distinct disease/pest classes. Open Access / CC BY-SA 4.0 license.
- **PlantDoc Dataset (IIT Delhi):** A benchmark dataset of 2,598 real-world field-condition photographs spanning 13 plant species and 17 pathological classes. Captures natural illumination variations, background occlusions, soil interference, and multi-leaf

scenes for domain robustness. Academic and Research Open Use.

- **Kaggle & ICAR Pedological Soil Collections:** Surface photography of Indian soil orders across diverse agricultural topographies (Alluvial plains, Deccan black cotton soil, coastal laterites, red sandy loams). Sampled from ICAR (Indian Council of Agricultural Research) regional institutes and open Kaggle pedological datasets.
- **Microsoft COCO & Pascal VOC Subsets (Non-Agricultural Distractors):** To train the Agricultural Relevance Gatekeeper, negative distractor images (motor vehicles, electronic consumer goods, interior rooms, selfies, processed fast foods) were sampled from Microsoft COCO and Pascal VOC benchmarks to ensure strict rejection of invalid uploads.
- **Alibaba Cloud Qwen 2.5-VL Multimodal Vision-Language Dataset:** Large-scale multimodal visual instruction and document tuning dataset powering Qwen2.5-VL:3B, enabling zero-shot symptom localization, anatomical plant part recognition (leaf, flower, fruit, stem, root), and agronomic reasoning.

3. Detailed Dataset Specifications

3.1. Agricultural Relevance Gatekeeper Dataset (agricultural_relevance_dataset.csv)

Purpose: Acts as the primary security and relevance filter at the API entry point. It evaluates uploaded images and immediately filters out non-agricultural inputs before they reach compute-intensive disease models.

Target Class	Samples	Description & Visual Archetype
plant	3,000	Crops, foliar leaves, botanical flowers, horticultural plants, vegetative canopy.
soil	3,000	Exposed agricultural topsoil, tilled fields, compost, seedbeds, potting soil.
non_agricultural	3,000	Automobiles, smartphones, human faces, furniture, architecture, processed snacks.

3.2. Crop Foliar Pathogen & Disease Dataset (crop_disease_dataset.csv)

Purpose: Multi-class foliar diagnostic model capable of recognizing 38 pathological conditions and healthy leaf stages across 14 economically critical agricultural crops (31,200 stratified records; 800 samples per class):

Crop Host	Supported Pathological / Healthy Classes	Count
Apple	Apple Scab (*Venturia inaequalis*), Black Rot (*Botryosphaeria obtusa*), Cedar Apple Rust (*Gymnosporangium*), Healthy	3,200
Blueberry	Healthy Foliage	800
Cherry	Powdery Mildew (*Podosphaera clandestina*), Healthy	1,600
Corn (Maize)	Cercospora Leaf Spot (Gray leaf spot), Common Rust (*Puccinia sorghi*), Northern Leaf Blight (*Exserohilum*), Healthy	3,200
Grape	Black Rot (*Guignardia bidwellii*), Esca (Black Measles), Leaf Blight (*Isariopsis clavispora*), Healthy	3,200
Orange / Citrus	Citrus Greening / Huanglongbing (*Candidatus Liberibacter*)	800
Peach	Bacterial Spot (*Xanthomonas campestris*), Healthy	1,600
Bell Pepper	Bacterial Spot (*Xanthomonas vesicatoria*), Healthy	1,600
Potato	Early Blight (*Alternaria solani*), Late Blight (*Phytophthora infestans*), Healthy	2,400
Raspberry	Healthy Foliage	800
Soybean	Healthy Foliage	800
Squash	Powdery Mildew (*Erysiphe cichoracearum*)	800
Strawberry	Leaf Scorch (*Diplocarpon earlianum*), Healthy	1,600
Tomato	Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Two-spotted Spider Mite, Target Spot, Tomato Yellow Leaf Curl Virus, Mosaic Virus, Healthy	8,000
Negative Control	General Agricultural Background / Soil	800

3.3. Botanical Plant Species Dataset (plant_species_dataset.csv)

Purpose: Multi-species visual recognition representing 98 distinct Indian agricultural, horticultural, and medicinal botanical species (14,700 samples; 150 per species) spanning cereals, pulses, vegetables, fruits, spices, trees, and floriculture:

Category	Key Representative Botanical Species	Species Count
Medicinal & Herbal	*Azadirachta indica* (Neem), *Ocimum tenuiflorum* (Tulsi), *Aloe barbadensis* (Aloe Vera), *Bacopa monnieri* (Brahmi), *Catharanthus roseus* (Sadabahar), *Mentha arvensis* (Mint), *Piper betle* (Paan)	18 species
Crops & Cereals	*Oryza sativa* (Paddy Rice), *Triticum aestivum* (Wheat), *Zea mays* (Maize/Corn), *Eleusine coracana* (Ragi), *Cicer arietinum* (Chickpea), *Cajanus cajan* (Pigeon Pea)	16 species
Vegetables	*Solanum lycopersicum* (Tomato), *Solanum melongena* (Brinjal), *Abelmoschus esculentus* (Okra/Bhindi), *Allium cepa* (Onion), *Allium sativum* (Garlic), *Capsicum annuum* (Chilli)	22 species
Fruit Crops	*Mangifera indica* (Mango), *Carica papaya* (Papaya), *Psidium guajava* (Guava), *Citrus limon* (Lemon), *Musa acuminata* (Banana), *Punica granatum* (Pomegranate), *Syzygium cumini* (Jamun)	20 species
Flowers & Ornamentals	*Rosa indica* (Rose), *Helianthus annuus* (Sunflower), *Hibiscus rosa-sinensis* (China Rose), *Jasminum sambac* (Mogra/Jasmine), *Tagetes erecta* (Marigold), *Nelumbo nucifera* (Lotus)	12 species
Agroforestry Trees	*Ficus religiosa* (Peepal), *Ficus benghalensis* (Banyan), *Cocos nucifera* (Coconut), *Moringa oleifera* (Drumstick), *Bambusa vulgaris* (Bamboo), *Eucalyptus globulus*	10 species

3.4. Pedological Soil Classification Dataset (soil_pedological_dataset.csv)

Purpose: Classifies agricultural surface soil into 6 major Indian pedological categories (9,000 samples; 1,500 samples per class) to determine irrigation capacity, soil organic carbon (SOC), and crop suitability:

Soil Category	Geographic Distribution in India	Agronomic Traits & Typical Crops	Samples
Alluvial Soil	Indo-Gangetic plains, Punjab, UP, Bihar, West Bengal, coastal deltas	High fertility, loamy texture, rich in potash/phosphoric acid. Ideal for Rice, Wheat, Sugarcane, Jute.	1,500
Black (Regur) Soil	Deccan plateau, Maharashtra, Gujarat, MP, northern Karnataka	High clay content, swelling/shrinkage self-ploughing, moisture retentive. Ideal for Cotton, Soybean, Sorghum.	1,500
Red Soil	Tamil Nadu, Karnataka, Andhra Pradesh, Odisha, Chota Nagpur	Rich in ferric oxides, porous, low nitrogen and humus. Ideal for Groundnut, Millets, Pulses, Tobacco.	1,500
Laterite Soil	Western Ghats summits, Kerala, coastal Karnataka, Assam hills	Intensely leached by heavy rainfall, acidic, high iron/aluminum. Ideal for Tea, Coffee, Cashew, Rubber.	1,500
Heavy Clay Soil	River backwaters, wetlands, lake beds, canal commanded tracts	Extremely fine grain, low drainage, high waterlogging risk. Ideal for Paddy Rice, Lotus, Water Chestnut.	1,500
Sandy Loam Soil	Rajasthan arid zones, south-west Haryana, coastal sand dunes	Coarse texture, high permeability, rapid leaching. Ideal for Bajra, Mustard, Watermelon, Guar.	1,500

4. 48-Dimensional Computer Vision Feature Engineering

Each tabular dataset row consists of 48 engineered numerical feature descriptors (**feat_0** through **feat_47**) extracted through specialized computer vision operations. These features enable ultralight (under 5 MB), ultra-fast (sub-50 millisecond) inference on standard CPUs without requiring expensive GPU hardware:

Feature Indices	Mathematical Extraction Method	Agronomic & Physiological Purpose
feat_0 – feat_2	**RGB Channel Color Moments:** First-order mean and variance across R, G, B channels after illumination normalization.	Separates dark foliage from light/glaucous leaves and distinguishes red/anthocyanin-rich vegetation.
feat_3 – feat_8	**Perceptual Color Spaces (HSV & CIELAB):** Hue angle circular mean, Saturation mean, Lightness L*, Green-Red chromaticity a*, Blue-Yellow b*.	Captures foliar chlorosis (yellowing), necrosis (browning), and diagnostic leaf pigmentation invariant to shadow.
feat_9 – feat_16	**Morphological Geometry:** Contour elongation aspect ratio (major/minor axis), circularity ($4\pi A/P^2$), solidity, compactness, convex hull ratio.	Differentiates linear grass blades (wheat/paddy) from broad ovate leaves (mango/guava) and lobed dicots (papaya).
feat_17 – feat_24	**Boundary Margin & Edge Density:** Sobel gradient magnitude histogram, Canny edge pixel density, and high-frequency boundary waviness.	Detects serrated margin teeth (rose/neem) vs smooth entire margins (banyan/mango) vs wavy undulating margins.
feat_25 – feat_36	**GLCM Texture & Venation Moments:** Gray-Level Co-occurrence Matrix Haralick descriptors (Energy, Contrast, Correlation, Homogeneity) at 0°, 45°, 90°.	Identifies parallel monocot venation vs reticulate dicot network, foliar pubescence (hairs), and surface smoothness.
feat_37 – feat_47	**Pathology & Symptom Ratios:** Calibrated pixel color thresholding for Chlorosis Index (% yellow), Necrosis Index (% brown), and Powdery Mildew Index (% white/gray).	Direct symptom quantifiers feeding the crop disease classifier and plant health score assessment algorithm.

5. Integrated Indian Botanical Database & Multimodal Vision LLM

In addition to the 4 CSV files, AGRO AI bundles two foundational knowledge bases to ensure accurate identification and agronomic management:

5.1. 10,208 Indian Botanical Species Catalog (Supabase PostgreSQL / SQLite)

A curated master taxonomy database covering India's agricultural, horticultural, and indigenous flora. Every species record contains 16 verified attributes: Common English Name, Scientific Binomial, Hindi Name, Telugu Name, Tamil Name, Botanical Family, Plant Category, Status in India (Native/Endemic/Introduced), Soil Preferences, Watering Requirements, Sunlight Tolerances, Growth Cycle, and Leaf Morphology JSON.

5.2. Qwen 2.5-VL 3B Multimodal Vision-Language Foundation Model

Integrated via local Ollama runtime (qwen2.5vl:3b). This 3-billion-parameter vision-language model performs deep visual screening on raw uncompressed image uploads. It analyzes organ type (flower, leaf, fruit, stem, root), evaluates ripeness and spoilage indicators, identifies visible foliar stress symptoms, and generates actionable, chemical-safe recommendations in structured JSON format.

6. Dataset Files, Models & Download Locations

All dataset CSV files, trained models, and reproduction scripts are persisted within your local AGRO AI installation and accessible at the following absolute paths:

Artifact / Dataset	Local File System Path	Format & Size
Agricultural Relevance Dataset	datasets/agricultural_relevance_dataset.csv	CSV (2.25 MB)
Crop Disease Dataset	datasets/crop_disease_dataset.csv	CSV (8.78 MB)
Plant Species Dataset	datasets/plant_species_dataset.csv	CSV (7.61 MB)
Soil Pedological Dataset	datasets/soil_pedological_dataset.csv	CSV (2.32 MB)
ONNX Plant Vision Model	backend/models/plant_identifier.onnx	ONNX (10.9 MB)
Joblib Ensemble Models	backend/models/{crop_disease, soil_classifier, agricultural_relevance}.joblib	Joblib (3 files)
Model Metadata Spec	backend/models/model_metadata.json	JSON Spec
Dataset Preparation Script	training/prepare_dataset.py	Python Script
Full Retraining Pipeline	training/train_all_models.py	Python Script

Reproducing and Retraining Models from Datasets:

```
# 1. Verify Dataset Integrity & File Formats:
python training/prepare_dataset.py

# 2. Train and export all models directly from CSV files:
python training/train_all_models.py

# 3. Verify Model Weights and Metadata Alignment:
python training/verify_trained_model.py
```

AGRO AI Engineering & Machine Learning Team
This document serves as the formal specification and reference manual for all datasets used in AGRO AI. For research questions, license verification, or dataset extensions, consult the project repository.